

Restaurant Channel Profitability Analysis for SkyCity Auckland Restaurants & Bars

1. Introduction

Restaurants today operate through multiple sales channels, including traditional in-store dining, third-party delivery platforms, and self-managed delivery services. While these channels increase customer reach and revenue opportunities, they also introduce different cost structures that directly affect profitability.

Revenue growth alone does not guarantee financial sustainability. Each sales channel incurs varying levels of food costs, operating expenses, commissions, and delivery-related expenses. As a result, channels generating high sales volumes may not necessarily generate high profits.

This study evaluates the profitability of four restaurant sales channels: In-Store Dining, Uber Eats, DoorDash, and Self-Delivery. The analysis focuses on understanding how costs impact profitability across channels and identifying which channels contribute most effectively to sustainable financial performance.

The project combines financial analysis, profitability assessment, cost decomposition, risk evaluation, and interactive dashboard development to support data-driven decision-making.

2. Problem Statement

Restaurant operators often focus on revenue and order volume when evaluating business performance. However, revenue alone provides limited insight into true profitability because each sales channel operates under a different cost structure.

Third-party delivery platforms introduce commission expenses, while self-delivery requires investment in logistics and transportation. Without a structured comparison of channel performance, restaurants may unintentionally prioritize channels that generate revenue but contribute little to overall profit.

The absence of channel-level profitability visibility makes it difficult to:

- Compare financial performance across channels.
- Understand the impact of commissions and delivery costs.
- Identify low-margin or loss-making channels.
- Make informed channel prioritization decisions.
- Evaluate the financial benefits of self-delivery operations.

This study addresses these challenges through a comprehensive profitability analysis of restaurant sales channels.

3. Objectives

Primary Objectives

- Compare net profitability across In-Store, Uber Eats, DoorDash, and Self-Delivery channels.

- Quantify the impact of commissions and delivery costs on profitability.
- Identify the most cost-efficient channel based on profit margins and profit per order.

Secondary Objectives

- Support data-driven channel prioritization.
 - Assess financial sensitivity to commission and delivery cost changes.
 - Compare profitability across cuisines and business segments.
 - Evaluate profit volatility and operational risk.
 - Develop an interactive dashboard for business stakeholders.
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4. Dataset Description

The dataset contains operational, revenue, cost, and profitability information for restaurant branches operating within the Auckland region.

Key variables include:

Restaurant Information

- RestaurantID
- RestaurantName
- CuisineType
- Segment
- Subregion

Operational Metrics

- GrowthFactor
- Average Order Value (AOV)
- MonthlyOrders
- InStoreOrders
- UberEatsOrders
- DoorDashOrders
- SelfDeliveryOrders

Revenue Metrics

- InStoreRevenue
- UberEatsRevenue
- DoorDashRevenue

- SelfDeliveryRevenue

Cost Metrics

- COGSRate
- OPEXRate
- CommissionRate
- DeliveryRadiusKM
- DeliveryCostPerOrder
- SD_DeliveryTotalCost

Profitability Metrics

- InStoreNetProfit
- UberEatsNetProfit
- DoorDashNetProfit
- SelfDeliveryNetProfit

Channel Share Metrics

- InStoreShare
- UE_share
- DD_share
- SD_share

Additional calculated fields were created during analysis, including:

- Channel-specific Cost of Goods Sold (COGS)
- Channel-specific Operating Expenses (OPEX)
- Aggregator Commission Costs
- Profit Per Order
- Margin Percentage
- Risk and Volatility Indicators

5. Methodology

The analysis was conducted in five stages.

5.1 Profit Calculation Validation

The provided profit formulas were validated by independently recalculating profits for each channel using revenue and cost components.

The following relationships were used:

$\text{InStore Profit} = \text{Revenue} - \text{COGS} - \text{OPEX}$

$\text{Uber Eats Profit} = \text{Revenue} - \text{COGS} - \text{OPEX} - \text{Commission}$

$\text{DoorDash Profit} = \text{Revenue} - \text{COGS} - \text{OPEX} - \text{Commission}$

$\text{Self-Delivery Profit} = \text{Revenue} - \text{COGS} - \text{OPEX} - \text{Delivery Cost}$

Differences between calculated and provided profits were examined to verify dataset consistency.

Profit values were also normalized on a per-order basis to enable fair comparison between channels with different order volumes.

5.2 Channel Margin Analysis

Profitability was evaluated using three primary indicators:

- Total Net Profit
- Profit Per Order
- Margin Percentage

These metrics allowed comparison of both total financial contribution and operational efficiency across channels.

5.3 Cost Component Decomposition

Revenue for each channel was decomposed into major cost components:

- Cost of Goods Sold (COGS)
- Operating Expenses (OPEX)
- Aggregator Commissions
- Delivery Logistics Costs

This analysis was used to identify the primary drivers of margin erosion.

5.4 Sensitivity Analysis

Scenario analysis was conducted to evaluate how profitability changes under varying cost assumptions.

Two factors were tested:

- Changes in commission rates for Uber Eats and DoorDash.
- Changes in self-delivery logistics costs.

The objective was to assess channel sensitivity and identify conditions under which profitability deteriorates.

5.5 Profitability and Risk Assessment

Profitability was further analyzed by cuisine type and business segment.

Margin variability was measured using standard deviation and relative risk indicators. Channels with high variability and frequent negative margins were identified as higher-risk operating models.

Loss-making channel occurrences were also evaluated to understand the prevalence of financially unsustainable strategies within the dataset.

6. Tools and Technologies Used

The analysis was performed using:

- Python
- Pandas
- NumPy
- Matplotlib
- Seaborn
- Plotly
- Streamlit

Python was used for data cleaning, transformation, statistical analysis, and profitability calculations. Streamlit was used to develop an interactive dashboard enabling real-time exploration of profitability metrics, cost structures, and scenario analysis.

7. Results and Findings

7.1 Profit Calculation Validation

The provided net profit values were independently recalculated using the revenue and cost components available in the dataset. Comparison between calculated profits and the provided profit values showed only negligible differences attributable to rounding.

No significant discrepancies were identified, indicating that the profitability calculations within the dataset are internally consistent and suitable for further analysis.

Profit values were also normalized on a per-order basis to enable comparison between channels with different order volumes.

Average Profit per Order

Channel	Profit per Order
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InStore	11.90
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Uber Eats	0.34
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DoorDash	0.34
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Self Delivery	8.78
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The results indicate substantial differences in channel efficiency. InStore generated the highest profit per order, followed by Self Delivery. Uber Eats and DoorDash produced significantly lower profit per order, suggesting that a large portion of revenue generated through aggregator platforms is absorbed by operating costs and commissions.

No extreme positive margin outliers were identified during validation. However, a number of restaurants recorded negative margins in delivery channels, indicating that profitability varies considerably across operating models.

7.2 Channel Margin Analysis

Total Net Profit by Channel

Channel	Total Net Profit
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InStore	3,828,417.36
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Uber Eats	258,488.08
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DoorDash	151,122.25
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Self Delivery	3,628,278.44
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InStore generated the highest total net profit within the dataset, closely followed by Self Delivery. Although Uber Eats and DoorDash generated substantial sales activity, their total profit contributions were significantly lower.

Average Margin Percentage

Channel	Average Margin (%)
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InStore	30.82
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Uber Eats	0.79
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DoorDash	0.79
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Self Delivery	22.60
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The margin analysis demonstrates that channel profitability differs substantially despite differences in revenue generation.

InStore achieved the strongest average margin at 30.82%, indicating the most efficient conversion of revenue into profit. Self Delivery maintained a healthy average margin of 22.60%, making it the second most profitable channel.

Both Uber Eats and DoorDash produced average margins below 1%, indicating that most revenue generated through these channels was consumed by operating expenses and commissions.

Interpretation

The results suggest that revenue volume alone is not an accurate indicator of channel performance. While delivery aggregators increase market reach and sales volume, their profitability remains substantially lower than InStore and Self Delivery operations within the analyzed dataset.

7.3 Cost Component Decomposition

To understand the causes of margin differences, revenue was decomposed into its major cost components.

Revenue and Cost Breakdown

Channel	Revenue	COGS	OPEX	Commission / Delivery Cost	Net Profit
InStore	14,284,378.07	4,226,913.99	6,229,046.81	0.00	3,828,417.36
Uber Eats	30,816,371.49	8,631,091.83	12,672,810.32	9,253,981.25	258,488.08
DoorDash	16,789,380.95	4,699,249.47	6,898,068.59	5,040,940.77	151,122.25
Self Delivery	15,849,176.29	4,428,545.49	6,510,542.83	1,281,809.47	3,628,278.44

Commission and Delivery Cost Impact

Average commission drag for both Uber Eats and DoorDash was approximately 30.03% of revenue.

Average self-delivery logistics impact was approximately 8.22% of revenue.

Interpretation

The cost decomposition reveals that commissions represent the primary source of margin erosion in aggregator channels.

Uber Eats generated the highest revenue among all channels. However, commission expenses alone exceeded 9.25 million, reducing overall profitability to only 258 thousand.

Similarly, DoorDash incurred more than 5 million in commission expenses, resulting in limited net profit despite substantial revenue generation.

Self Delivery incurred logistics costs, but these costs were considerably lower than aggregator commissions. As a result, Self Delivery maintained significantly stronger profitability than both Uber Eats and DoorDash.

7.4 Commission and Delivery Cost Sensitivity Analysis

Sensitivity analysis was performed to evaluate how profitability changes under varying cost conditions.

Commission Impact

Average commission per order:

Platform Commission per Order

Uber Eats 11.56

DoorDash 11.56

Simulation results showed that increasing commission rates caused rapid declines in aggregator profitability. Under higher commission scenarios, channel profits decreased substantially and eventually became negative.

This indicates that aggregator profitability is highly sensitive to commission structures.

Delivery Cost Impact

Average self-delivery cost per order:

Channel Cost per Order

Self Delivery 3.12

Simulation results showed that increasing delivery costs reduced Self Delivery profitability. However, Self Delivery remained profitable across a broader range of cost scenarios compared to aggregator channels.

Interpretation

The analysis suggests that aggregator profitability depends heavily on commission rates, while Self Delivery demonstrates greater resilience to cost increases.

Within the analyzed dataset, delivery logistics costs remained considerably lower than aggregator commission expenses.

7.5 Cuisine-wise Profitability Analysis

Average channel margins were calculated for each cuisine category.

Average Margin by Cuisine (%)

Cuisine	InStore	Uber Eats	DoorDash	Self Delivery
Burgers	31.02	1.04	1.04	22.76
Chicken Dishes	28.85	-1.36	-1.36	20.35
Chinese	31.02	0.95	0.95	22.99
Indian	31.41	1.51	1.51	23.32

Cuisine	InStore	Uber Eats	DoorDash	Self Delivery
Japanese	31.83	1.93	1.93	24.39
Kebabs/Mediterranean	29.74	-0.29	-0.29	21.02
Pizza	30.39	0.27	0.27	22.15
Thai	31.94	1.76	1.76	23.38

Interpretation

Japanese, Thai, and Indian cuisines recorded the highest average margins across most channels within the dataset.

Chicken Dishes and Kebabs/Mediterranean recorded the lowest margins and were the only cuisine groups with negative average margins on aggregator platforms.

These results suggest that cuisine characteristics may influence the ability to absorb delivery-related costs and commissions.

7.6 Segment-wise Profitability Analysis

Average margins were also analyzed by business segment.

Average Margin by Segment (%)

Segment	InStore	Uber Eats	DoorDash	Self Delivery
Cafe	36.12	6.12	6.12	27.96
Full-service	12.53	-17.44	-17.44	4.18
Ghost Kitchen	45.28	15.11	15.11	36.81
QSR	35.88	5.83	5.83	27.80

Interpretation

Ghost Kitchens recorded the highest average margins across all channels in the dataset.

Full-service restaurants recorded the lowest margins and experienced negative average margins on both Uber Eats and DoorDash.

This suggests that business models with lower operating overheads may be better positioned to maintain profitability across delivery channels.

7.7 Profit Volatility and Risk Assessment

Margin variability was measured using standard deviation.

Margin Volatility

Channel	Volatility
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InStore	11.99
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Uber Eats	12.07
-----------	-------

DoorDash	12.07
----------	-------

Self Delivery	12.59
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Risk Index

Channel	Risk Index
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InStore	0.39
---------	------

Uber Eats	15.23
-----------	-------

DoorDash	15.23
----------	-------

Self Delivery	0.56
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The risk index was calculated as the ratio of margin volatility to average margin.

Loss-Making Channel Occurrences

Channel	Restaurants with Negative Margin
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InStore	0
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Uber Eats	538
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DoorDash	538
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Self Delivery	90
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Interpretation

InStore recorded no loss-making cases within the dataset and maintained the lowest risk index.

Uber Eats and DoorDash recorded the highest risk indices due to their extremely low average margins combined with relatively high variability.

More than 500 restaurant-channel combinations produced negative margins on both aggregator platforms, compared with only 90 loss-making cases for Self Delivery.

These findings indicate that aggregator profitability is considerably less stable than InStore and Self Delivery operations within the analyzed dataset.

8. Dashboard Development

An interactive Streamlit dashboard was developed to support exploratory analysis and decision-making. The dashboard enables users to evaluate profitability across restaurant sales channels and assess the impact of different cost structures.

The dashboard was designed around the core objectives of the study and provides a visual interface for comparing channel performance, identifying cost drivers, and conducting scenario analysis.

Key Dashboard Modules

Channel-wise Profit Comparison

This module compares total net profit across InStore, Uber Eats, DoorDash, and Self Delivery channels. Users can quickly identify which channels contribute the most to overall profitability and compare profit distribution across the business.

Margin Analysis

The margin analysis module displays average margin percentages and profitability indicators for each channel. This allows users to compare operational efficiency rather than focusing solely on revenue or total profit.

Cost Breakdown and Waterfall Analysis

The cost breakdown module decomposes channel revenue into major cost components including:

- Cost of Goods Sold (COGS)
- Operating Expenses (OPEX)
- Aggregator Commissions
- Self-Delivery Logistics Costs

Waterfall charts visually illustrate how revenue is transformed into net profit after successive cost deductions. This provides a clear view of where margin erosion occurs within each channel.

Cuisine and Segment Profitability Analysis

Heatmaps were developed to compare margin performance across:

- Cuisine Types
- Business Segments

These visualizations allow users to identify categories with stronger or weaker profitability profiles and evaluate whether channel performance differs across restaurant formats.

What-If Analysis

Interactive sliders enable users to simulate changes in:

- Aggregator Commission Rates
- Self-Delivery Logistics Costs

The resulting profit estimates help users understand how sensitive channel profitability is to changes in cost structures and support scenario-based planning.

User Filters

The dashboard includes filtering options for:

- Cuisine Type

- Business Segment
- Sales Channel

These filters allow stakeholders to focus on specific restaurant categories and investigate profitability patterns at a more granular level.

Business Value

The dashboard converts raw operational and financial data into an interactive decision-support system. Rather than relying on static reports, users can dynamically explore profitability drivers, compare channels, and evaluate potential strategic adjustments.

9. Recommendations

The recommendations presented below are based solely on the results obtained from the analyzed dataset.

9.1 Prioritize High-Margin Channels

The analysis found that InStore operations generated the highest average margin (30.82%) and the highest total net profit (3.83 million).

Self Delivery generated the second-highest average margin (22.60%) and produced total profits nearly equal to InStore operations.

Based on these findings, restaurants may benefit from prioritizing channel strategies that strengthen InStore and Self Delivery performance.

9.2 Monitor Aggregator Dependence

Uber Eats and DoorDash generated average margins below 1% despite producing substantial revenue.

The analysis showed that commission expenses represented approximately 30% of revenue and significantly reduced channel profitability.

Restaurants with heavy dependence on aggregator platforms should regularly evaluate whether revenue growth is translating into meaningful profit growth.

9.3 Evaluate Commission Structures

Sensitivity analysis demonstrated that aggregator profitability is highly responsive to commission rate changes.

As commission expenses increase, channel profitability declines rapidly and can become negative under certain conditions.

Businesses may benefit from:

- Reviewing commission agreements

- Negotiating lower commission rates where possible
 - Evaluating the profitability of promotional campaigns offered through aggregators
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9.4 Expand Self-Delivery Where Economically Feasible

The dataset showed that self-delivery logistics costs averaged approximately 8.22% of revenue, substantially lower than aggregator commission costs.

Although self-delivery requires operational resources, the results suggest that it may provide stronger profit retention compared with third-party delivery platforms.

Restaurants should evaluate whether expanding self-delivery capabilities is economically viable within their operating environment.

9.5 Apply Segment-Specific Channel Strategies

The analysis revealed substantial differences across business segments.

Ghost Kitchens recorded the strongest margins across all channels, while Full-Service restaurants recorded the weakest performance and negative average margins on aggregator platforms.

These findings suggest that channel strategies should be tailored to business models rather than applied uniformly across all restaurant types.

9.6 Monitor Loss-Making Channels

The study identified:

- 538 negative-margin Uber Eats cases
- 538 negative-margin DoorDash cases
- 90 negative-margin Self Delivery cases
- 0 negative-margin InStore cases

Restaurants should routinely monitor channel-level profitability and identify channels that consistently operate at a loss.

Continued investment in loss-making channels may reduce overall business profitability despite generating additional revenue.

9.7 Use Profitability Metrics Alongside Revenue Metrics

The analysis demonstrates that high revenue does not necessarily translate into high profitability.

Future performance monitoring should include:

- Net Profit

- Margin Percentage
- Profit per Order
- Cost Composition
- Risk Indicators

alongside traditional revenue-based measures.

This approach provides a more complete understanding of business performance and supports sustainable decision-making.

9.8 Limitations of the Study

While the analysis provides valuable insights into channel profitability, several limitations should be considered when interpreting the results.

Dataset Scope

The findings are based entirely on the provided dataset and therefore reflect the characteristics and assumptions contained within that dataset. Results should be interpreted within the context of the analyzed data rather than generalized to all restaurants or delivery platforms.

Aggregated Financial Measures

The analysis relies on aggregated monthly revenue, cost, and profit figures. Individual order-level factors such as promotional discounts, customer acquisition costs, peak-hour pricing, and order-specific operational variations were not available for evaluation.

Limited Cost Variables

Profitability calculations were based on the cost variables provided in the dataset, including COGS, OPEX, commissions, and delivery costs. Other potential business expenses such as rent, marketing expenditure, labor allocation, technology fees, taxes, and capital investments were not explicitly included in the analysis.

Scenario Analysis Assumptions

The what-if analysis assumes that commission rates and delivery costs can change independently while all other business variables remain constant. In practice, changes in pricing structures may also influence customer demand, order volume, and operational behavior.

Geographic Context

The dataset represents restaurant operations within Auckland and may not reflect market conditions, customer behavior, cost structures, or competitive environments in other regions.

Operational Factors Not Included

The analysis focuses primarily on financial performance. Non-financial considerations such as customer acquisition, brand visibility, customer retention, delivery reliability, and strategic market expansion were outside the scope of this study.

Future Research Opportunities

Future studies could incorporate:

- Customer satisfaction metrics
- Order-level transaction data
- Marketing and promotional expenses
- Dynamic pricing effects
- Longitudinal performance trends across multiple time periods
- Predictive profitability forecasting models

Incorporating these factors would provide a more comprehensive understanding of restaurant channel performance and long-term business sustainability.

10. Conclusion

This study examined the profitability of four restaurant sales channels: InStore, Uber Eats, DoorDash, and Self Delivery.

The results demonstrated significant differences in profitability despite substantial variations in revenue generation across channels.

InStore operations generated the highest average margin (30.82%) and the highest total net profit within the dataset. Self Delivery emerged as the second strongest channel, producing margins of 22.60% and generating profits comparable to InStore operations.

Although Uber Eats and DoorDash generated substantial revenue, average margins remained below 1%. Cost decomposition analysis identified aggregator commissions as a major contributor to margin erosion, significantly reducing the profitability of these channels.

Cuisine-level analysis showed variability in profitability across restaurant categories, while segment-level analysis revealed notable differences between business models. Ghost Kitchens recorded the strongest margins across channels, whereas Full-Service restaurants experienced the weakest profitability performance.

Risk assessment further indicated that aggregator channels exhibited substantially higher financial risk than InStore and Self Delivery operations. More than 500 restaurant-channel combinations recorded negative margins on both Uber Eats and DoorDash, compared with 90 cases for Self Delivery and none for InStore operations.

The Streamlit dashboard developed as part of this project provides an interactive framework for evaluating channel performance, analyzing cost structures, and conducting profitability simulations.

Overall, the findings reinforce the importance of evaluating channels based on profitability rather than revenue alone. Within the analyzed dataset, InStore and Self Delivery channels demonstrated stronger financial performance, while aggregator channels exhibited significantly greater exposure to commission-driven margin erosion. These insights can support more informed channel management and profitability-focused decision-making.